2026 Data Visualisation Project Brief

Hospitalised Injuries from Road Crashes in Australia

Background

Road traffic accidents represent a significant public health issue in Australia. They are consistently identified as **one of the leading causes of injury-related hospitalisation nationally (https://www.aihw.gov.au/reports/injury/injury-in-australia/contents/ranked-causes)** . Each year, tens of thousands of individuals are admitted to hospital due to injuries sustained in road traffic accidents. While fatalities often receive substantial media and policy attention, non-fatal injuries constitute a considerably larger and ongoing burden on individuals, families, communities, and the healthcare system.

Australian government agencies publish detailed, open-access datasets documenting hospitalised injuries resulting from road traffic accidents. These datasets include injury counts by year, jurisdiction, demographic characteristics, road user type, and selected population groups.

Project Aim

The aim of this project is to critically analyse and visualise trends related to hospitalised injuries from road traffic accidents in Australia.

Students are expected to:

- Conduct independent exploration of the dataset
- Identify and articulate substantively meaningful patterns, trends, or contrasts
- Design and implement visualisations that effectively communicate analytical insights
- Construct a structured narrative supported by empirical evidence and presented as an interactive web visualisation.

This project emphasises data processing, exploration and analysis and purposeful visual design. It is not intended to produce an unconstrained exploratory dashboard enabling users to view every possible data combination. Rather, the focus is on developing a data story supported by appropriate visual representations.

Team Design Task

Working in teams of two, you will design and implement one or more interactive data visualisations intended for a web-based audience. Your visualisation(s) must:

- Present a clear and analytically grounded narrative about hospitalised road crash injuries

- Enable users to explore selected dimensions of the dataset in a structured and meaningful way
- Demonstrate considered design decisions regarding layout, interaction, and visual encoding, and
- Support both exploratory analysis and explanatory communication.

In addition, your team is required to:

- document your data processing and design process
- conduct and document a usability evaluation of your visualisation using appropriate usability testing methods.

You may draw on relevant domain literature and supplementary datasets to contextualise and interpret observed patterns. However, the primary emphasis of this assessment is on identifying, analysing, and communicating meaningful insights grounded in the hospitalisation data itself.

Dataset

The primary dataset for this project is as follows:

> Bureau of Infrastructure and Transport Research Economics (BITRE) Hospitalised Injury Data:
>
> **https://www.bitre.gov.au/publications/ongoing/hospitalised-injury (https://www.bitre.gov.au/publications/ongoing/hospitalised-injury)**

The dataset comprises:

- National hospitalised injury counts (2011–2021)
- State and Territory hospitalised injury counts (2011–2021)
- First Nations Peoples hospitalised injury counts (2011–2021)

> An accompanying Data Dictionary

Although summary tables are provided in the data files, you are strongly encouraged to derive your own aggregations and transformations directly from the raw data where appropriate (see last tab of each data file).

BRITE has developed a Power BI application ("**Hospitalised injuries from road crashes (https://app.powerbi.com/view?r=eyJrIjoiZWQ0NTQ3ZDMtYWM1ZS00ZTNhLTlmZmMtZWY1MWFjOWNjODc5IiwidCI6ImFhMjFiNjQwLWJhYzItNDU2ZC04NTA1LWYyY2MwN2Y1MTc4NCJ9)** ") to help users explore some aspects of the data set. You will be required to undertake a critical analysis of this dashboard for the Critique Assignment. However, this application should be treated as contextual reference only.

Its structure and scope are limited, and replication of its design or purpose is neither required nor encouraged in your project.

Note on Interpreting Count Data

The BITRE dataset reports absolute counts of hospitalised injuries. Direct comparison of raw counts across jurisdictions, demographic groups etc may be misleading where substantial differences in population size exist (e.g., Northern Territory compared with Victoria). You are not required to normalise this data to pass this assessment. However, it must be clear in your data story narrative that you are aware of it's limitations.

Project Resources

Personnel

For this Project you will work in a team of 2 students. It is expected that your team will be quite diverse, with students specialising in software, information systems, networking, data science and cybersecurity participating in this unit. In addition, there a variety of nationalities and educational backgrounds. This diversity reflects the real-world nature of work in a modern world.

Time

Teams will be formed in Week 8 (after the Easter semester break). The team will have approximately 5 weeks in total to complete the project with submissions due just after the end of semester.

It is expected that each student will need to put in approximately 6-8 hours of work per week on this project in the second half of semester.

Software

- Visual Studio Code (or code editor of your choice)
- GitHub
- KNIME
- Confluence (optional)
- Trello (optional)

Web hosting

Swinburne Mercury server

Technical Skills

Most students in this unit should have completed First and Second Year or the equivalent in either:

- Bachelor of Information and Communication Technology,
- Bachelor of Cybersecurity, or
- Bachelor of Computer Science (Data Science major).

Minimum skill:

COS10009 Introduction to Programming is a requisite for this unit.

Other skills:

Unless doing this unit as an elective in degree not listed above, students should also have skills in:

- Basic web programming (e.g., COS10005, COS10026)
- Teamwork and team management (e.g. ICT10022, ICT20025, COS10025, COS10026, COS20031).

Other relevant skills your team members may have:

- Design thinking and usability testing (e.g., ICT20025)
- KNIME (e.g., COS10022)
- Database (e.g., COS20026, COS20015)

Students who completed ICT20025 and COS20031 are expected to be familiar with Jira and Confluence.

D3 Skills

The visualisations in your website must be built using D3.

The programming exercises in the Class component of the unit will equip you to import simple data sets, construct basic chart types and simple interactive features in D3.

However, for more complicated/custom charts and data sets necessary for high marks in some project assessment criteria your team will need to undertake further independent research.

Generic Skills

To complete this Project you will need to have (or develop) the following generic skills:

- good English written and spoken communication
- good organisational and time management skills
- basic word processing skills, including use document templates (style headings etc)
- excel skills

- Swinburne holds a number of study, academic and language workshops to help you improve on these basic skills:
  - **Workshops and short courses, English Language Skills (http://www.swinburne.edu.au/student/study-help/las/workshops.html)**
  - **Study Support and Academic Skills (https://www.swinburne.edu.au/current-students/study-support/)**

These workshops and skills sessions are free for Swinburne Students.

Skill Level

We understand that there will be students who are stronger in some skills than others. Where possible we will encourage the formation of balanced teams so each team member has a chance to contribute to all aspects of the project.